Remarks on Topological Strings

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January 2020

Abstract

This paper proves a local large-N Koszul-duality theorem for the Hamiltonian sector of the mixed A/B topological string on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$. The ghost-zero open BV theory reduces to the gauged first-order matrix model

$$\int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

After scalar reduction and degreewise stabilization, its Hamiltonian algebra is

$$\mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big), \qquad \bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1,$$

where $\bar{A}_{\mathrm{desc}}$ is represented by the nonconstant primitive one-antifield Koszul cycles. The closed Hamiltonian sector is cotangent BF theory for the Tate Lie algebra

$$\mathfrak{g} = (z_1, z_2)\mathbb{C}[[z_1, z_2]] \ltimes \big((z_1, z_2)\mathbb{C}[[z_1, z_2]]\big)^{\vee}_{\mathrm{cont}}[1].$$

Its local observables are the continuous Chevalley-Eilenberg cochains $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g})$, and their Koszul dual is the completed enveloping algebra $\widehat{U}(\mathfrak{g})$. The PBW special fibre identifies degree by degree with the stable open Hamiltonian algebra. Boundary evaluation

$$f(z_1, z_2) \longmapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$$

identifies the degree-zero generators and intertwines the closed Hamiltonian Poisson bracket with the scalar-reduced open trace bracket.

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0.1 Introduction

0.1.1 Overview

Koszul duality is the organizing algebraic mechanism for the local twisted M-theory model studied here.

Remark 0.1.1. This introduction records the physical source of the local theorem. The mathematical assertions are the formal local statements proved below.

Let X be a holomorphic symplectic manifold. Consider a topological twist of M-theory on $\mathbb{R} \times X$ in the presence of a stack of N one-dimensional topological branes supported on $\mathbb{R} \times p$, for some point $p \in X$.

The local model contains two algebras:

- The algebra $\mathcal{A}_N$ of operators on the stack of N branes supported on $\mathbb{R} \times p$.
- The algebra $\mathcal{B}$ of local operators of the gravitational theory on $\mathbb{R} \times X$. Operator product expansions in the direction of $\mathbb{R}$ equip $\mathcal{B}$ with the structure of an associative algebra.

The guiding large-N statement is:

Conjecture 0.1.2. As $N \to \infty$, $\mathcal{A}_N$ is the Koszul dual of $\mathcal{B}$.

Remark 0.1.3. Witten's Chern-Simons model [4] supplies the open-string algebra used for operator-product calculations.

These notes isolate one local mixed A/B example and verify the classical stable operator-algebra match. The quantum discussion is restricted to the canonical first-order bracket and the formal odd-power Moyal targets. The closed-string Feynman integral construction is formulated as Problem 0.2.12.

0.1.2 *The local theorem*

Let

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1, \qquad \widehat{\mathfrak{h}} = (z_1, z_2)\mathbb{C}[[z_1, z_2]], \qquad \mathfrak{h} = \widehat{\mathfrak{h}}, \qquad \mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}},$$

and set

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee[1],$$

where $\mathfrak{h}$ carries the Poisson bracket

$$\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g,$$

understood after projection modulo constants, and $\mathfrak{h}^\vee$ has the continuous dual Taylor basis $\delta_{k,l}$ for $k + l > 0$. We use $\bar{A}$ for degreewise polynomial calculations and $\widehat{\mathfrak{h}}$ for the completed closed CE/Koszul algebra. Constants are removed because a constant Hamiltonian has zero Hamiltonian vector field. On the open side $\mathrm{Tr}(1) = N$ is a scalar parameter. The symbol $\bar{A}_{\mathrm{desc}}$ denotes an abstract copy of $\bar{A}$ whose nonconstant basis vectors are represented by the first descendant cycles constructed below.

Theorem 0.1.4 *(Local stable Hamiltonian-sector Koszul duality).* In the formal local mixed A/B model on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$, the stable large-N scalar-reduced open Hamiltonian algebra freely generated by the primitive degree-zero Hamiltonian traces and the nonconstant primitive one-antifield Koszul classes is

$$O^{\mathrm{cl}}_{\mathrm{brane}} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big).$$

The classical local operators in the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg algebra

$$O^{\mathrm{cl}}_{\mathrm{grav}} \cong C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

Its Koszul dual is the completed graded enveloping algebra

$$\big(O^{\mathrm{cl}}_{\mathrm{grav}}\big)^! \simeq \widehat{U(\mathfrak{g})}.$$

The PBW/order Rees algebra $U_\hbar(\mathfrak{g})$ has completed special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

Its degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$ identifies with the open stable Hamiltonian-sector algebra after choosing the monomial-dual basis $\mathfrak{h}^\vee \cong \bar{A}_{\mathrm{desc}}$. This is a PBW special-fibre product identification; the open-side deformation is checked below on degree-zero Hamiltonian generators. On those generators the identification is induced by the scalar-reduced bulk-to-boundary evaluation map

$$\partial_{\mathrm{bb}}(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)},$$

which sends a closed Hamiltonian to the corresponding brane Hamiltonian and intertwines Poisson brackets. The shifted coadjoint generators are sent to the nonconstant first Koszul descendants $\Psi_{k,l}$. On monomials,

$$z_1^k z_2^l \longleftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \qquad k + l > 0,$$

the first bracket is

$$\{\overline{\mathrm{Tr}(\phi_1^k\phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m\phi_2^n)}\} = \overline{(kn-lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})},$$

matching the Poisson bracket on $\bar{A}$. If a displayed exponent is negative, the coefficient is already zero; if the displayed monomial is constant, scalar reduction kills it.

0.1.3 *Status of assertions*

The theorem above is the precise local assertion proved in these notes; the terms appearing in it are constructed in the subsections below. The proof consists of the affine Ext calculation, the open BV reduction, the stable invariant-theory calculation of the scalar-reduced Hamiltonian trace algebra, the computation of the primitive one-antifield Koszul homology, the formal Hamiltonian BF model for the closed side, the construction of the bulk-to-boundary Hamiltonian evaluation map, the CE/Koszul-duality computation, and the first bracket calculation.

The formal Weyl/Moyal calculation below is an all-order algebraic deformation-quantization target for the constant Poisson formal disk; its monomial coefficients are checked by an exact script. A closed-string Feynman-integral derivation of those coefficients is a separate analytic problem. It requires the global BV/gauge-fixing data listed in Problem 0.2.12. The proof of the local theorem uses the formal local model alone.

The descendant generators are identified with $\mathfrak{h}^\vee[1]$ in the PBW special fibre. Proposition 0.2.38 computes the open Poisson action on polynomial descendants, and Corollary 0.2.39 identifies the coadjoint action as a different representation. The polynomial boundary model lacks a deformation-level descendant match.

0.1.4 *Conventions for the local calculation*

Everything below is over $\mathbb{C}$. The phrase *formal local model* means the completed Taylor model at the chosen point of $\mathbb{C}^2$; equivalently, one may work polynomially degree by degree and then complete at the augmentation ideal. The symbol $\widehat{\mathbf{S}}$ denotes this completed symmetric algebra, and all Chevalley-Eilenberg cochains are continuous cochains for the same filtration.

Shifts follow the BV convention used in the field descriptions: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of V. The notation $\mathbf{S}(V)$ always means the free graded-commutative algebra on the graded vector space V.

We write $\bar{A}_{\mathrm{desc}}$ for an abstract copy of $\bar{A}$ whose basis vector labelled by $z_1^k z_2^l$, with $k+l>0$, is represented on the brane by the first descendant $\Psi_{k,l}$. On the closed side the cotangent field gives the continuous Taylor dual $\mathfrak{h}^\vee$. Choosing the monomial Taylor basis identifies $\mathfrak{h}^\vee$ with $\bar{A}_{\mathrm{desc}}$ as a filtered vector space. The closed module structure remains the coadjoint module structure.

Boundary terms on $\mathbb{R}$ are discarded by taking compactly supported variations, or equivalently by working on an interval with fixed boundary values and then passing to local operators in the interior. The term *stable large-N* means the degreewise stable value of the invariant algebras under block restriction as $N \to \infty$. Inside a fixed finite matrix algebra all finite-N trace identities remain present. The scalar trace $\mathrm{Tr}(1) = N$ is treated as a parameter and is removed from the reduced operator algebra. If T is a trace expression, $\overline{T}$ denotes its class modulo the scalar line $\mathbb{C}\cdot\mathrm{Tr}(1)$.

On the open side we compute the scalar-reduced primitive degree-zero Hamiltonian traces and the nonconstant primitive one-ψ Koszul homology. The scalar class $\mathrm{Tr}(\psi)$ is computed by the same finite argument and excluded from the Hamiltonian comparison, since constants have been removed from $\bar{A}$. Higher ψ-count homology lies outside the Hamiltonian comparison.

0.1.5 *The concrete dictionary*

The comparison is the following explicit dictionary. For $k, l \geq 0$ and $k + l > 0$, set

$$H_{k,l} = z_1^k z_2^l \in \bar{A}, \qquad E_{k,l} = \delta_{k,l}[1] \in \mathfrak{h}^\vee[1],$$

where $\delta_{k,l}$ is dual to the Taylor monomial $z_1^k z_2^l$. On the brane side set

$$O_{k,l} = \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)},$$

and let $\Psi_{k,l}$ be the symmetrised trace obtained by inserting one ψ into the word with k copies of ϕ_1 and l copies of ϕ_2; the odd matrix ψ is the Koszul variable for the moment-map equation, introduced in Construction 0.2.17. The theorem proves that the assignment

$$H_{k,l} \longmapsto O_{k,l}, \qquad E_{k,l} \longmapsto \Psi_{k,l}$$

extends to an isomorphism of the scalar-reduced Hamiltonian descendant algebra with

$$\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]) \xrightarrow{\sim} \mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}}$$

in the sector considered here. On the closed side the symbols $H_{k,l}, E_{k,l}$ form the degreewise polynomial PBW generators inside the special fibre of $U_\hbar(\mathfrak{g})$.

Thus every later step has a concrete target: construct the symbols $O_{k,l}, \Psi_{k,l}$ from the open fields, construct $H_{k,l}, E_{k,l}$ from Hamiltonians and cotangent Hamiltonians, prove that commutators of words reduce to commutative monomials, and check that the bracket coefficient is $(kn - lm)$.

0.1.6 *String Field Theory and Koszul Duality*

We expect Koszul duality for string field theories to be reflected in stable trace invariants and cyclic homology. The algebraic template is the Loday-Quillen-Tsygan theorem [2]tsygan:

Theorem 0.1.5 *(Loday-Quillen-Tsygan).* Let A be a unital associative algebra over a field of characteristic zero. The stable Chevalley-Eilenberg homology of $\mathfrak{gl}_\infty(A)$ is the free graded-commutative algebra generated by the shifted cyclic homology of A:

$$H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A)\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

Thus stable matrix invariants are controlled by cyclic words. In the particular reduced model below, BRST exactness of $[\phi_1, \phi_2]$ turns cyclic words into ordinary commutative monomials, and the surviving single-trace generators are represented by $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$.

The calculation below is the corresponding classical shadow: closed Hamiltonian modes match stable single-trace generators, and the first nontrivial bracket agrees with the Poisson bracket.

0.1.7 *Flavours of Topological Strings*

Topological strings manifest in various flavours. Common examples include:

- The **A-model**, where branes are objects of a Fukaya category: Lagrangians equipped with the grading, spin or Pin, local-system, and unobstructedness data required to define Floer theory.
- The **B-model**, where branes are objects of $D^b\mathrm{Coh}(X)$, or of $\mathrm{Perf}(X)$ in the affine local model.

- **Mixed A-B models**, e.g. on $X \times X'$ with branes given by $L \times Z$ for $L \rightarrowtail X$ Lagrangian and $Z \rightarrowtail X'$ holomorphic.

Remark 0.1.6. The enumerative content of the A-model includes its world-sheet instanton sectors.

The construction has two structural inputs:

- The open string theory, which is the theory on a brane, and
- The closed string field theory, of a gravitational nature.

0.1.8 *Open Strings*

If $\mathcal{C}$ is a cyclic Calabi-Yau A_∞ category and $\mathcal{F} \in \operatorname{Ob}\mathcal{C}$, set $A = \operatorname{RHom}(\mathcal{F}, \mathcal{F})$. The open-string states are A. The BV fields are[1]

$$A[1].$$

The Calabi-Yau structure supplies a cyclic trace $\operatorname{Tr} : A \to \mathbb{C}[-d]$. For a cyclic A_∞ algebra with operations m_n, the formal Chern-Simons action is

$$S(a) = \sum_{n \geq 1} \frac{1}{n+1} \operatorname{Tr}\big(a\, m_n(a, \ldots, a)\big), \qquad a \in A[1].$$

This is the algebraic core of Witten's open string field theory [4] and of the AKSZ packaging [5].

0.2 The Local Model

0.2.1 *Setup*

Let $X = \mathbb{R}^2_A \times \mathbb{C}^2_B$. We use the exact A-brane given by the standard line $\mathbb{R} \subset \mathbb{R}^2_A$ with trivial brane data and the B-brane $\mathcal{O}_p \in \operatorname{Perf}(\mathbb{C}^2_B)$. Consider the product brane

$$\mathbb{R} \times p \rightarrowtail \mathbb{R}^2_A \times \mathbb{C}^2_B$$

for some point $p \in \mathbb{C}^2_B$. The A-model lives on $\mathbb{R}^2_A$ (equipped with its standard symplectic structure) and the B-model lives on $\mathbb{C}^2_B$ (equipped with its standard holomorphic-symplectic structure $\omega = dz_1 \wedge dz_2$).

0.2.2 *Open String Field States*

The open string states are the tensor product of A-model and B-model states.

- *A-model factor.* In the exact self-Floer model for $\mathbb{R} \subset T^*\mathbb{R} \cong \mathbb{R}^2$, a small Hamiltonian perturbation identifies $CF^*(\mathbb{R}, \mathbb{R})$ with the Morse complex of $\mathbb{R}$. Hence $HF^*(\mathbb{R}, \mathbb{R}) \cong H^*(\mathbb{R}; \mathbb{C})$, and the de Rham complex $\Omega^*(\mathbb{R})$ is the chain-level model used below.
- *B-model factor.* Put affine coordinates at $p = 0$ and write $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $\mathcal{O}_p = R/\mathfrak{m}$. The skyscraper sheaf has open string states

$$\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m}),$$

  computed in Proposition 0.2.1.

[1] If we produce a category, the resulting field theory is defined on a point. If instead we produce a sheaf of categories, we get a field theory on the support of this sheaf, i.e. a field theory on the given brane.

Proposition 0.2.1 *(Affine skyscraper self-Ext).* Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. Then the Yoneda algebra of the skyscraper sheaf at the origin is

$$\mathbf{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2),$$

where $|\varepsilon_i| = 1$. In particular this is the free graded-commutative algebra on two odd generators.

Proof. The pair (z_1, z_2) is a regular sequence in the polynomial ring R, so the affine Koszul complex

$$K = \left(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\right)$$

is a free resolution of k, with

$$d_1(ae_1 + be_2) = az_1 + bz_2, \qquad d_2(e_{12}) = z_2 e_1 - z_1 e_2.$$

Applying $\mathrm{Hom}_R(-, k)$ gives

$$0 \to k \xrightarrow{0} k\, \varepsilon_1 \oplus k\, \varepsilon_2 \xrightarrow{0} k\, \varepsilon_{12} \to 0,$$

because z_1 and z_2 act by zero on k. Hence the graded vector space of Ext groups is k in degree 0, $k\varepsilon_1 \oplus k\varepsilon_2$ in degree 1, and $k\varepsilon_{12}$ in degree 2.

The product is the Yoneda product. The Koszul resolution is the exterior dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$; its diagonal sends e_i to $e_i \otimes 1 + 1 \otimes e_i$. Dualising this diagonal gives $\varepsilon_i^2 = 0$ and $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, while $\varepsilon_1\varepsilon_2$ spans the top Ext group. Thus the Yoneda algebra is $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$. □

0.2.3 *Open String Field Theory*

The total space of fields (in the Batalin-Vilkovisky formalism) is

$$\alpha \in \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$$

with action

$$S(\alpha) = \int_{\mathbb{R}\times\mathbb{C}^{0|2}} \frac{1}{2}\,\mathrm{Tr}(\alpha\, d\alpha) + \frac{1}{3}\,\mathrm{Tr}\,\alpha^3.$$

Here $d = d_{\mathrm{dR}}$; the Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ has zero differential. The integral over $\mathbb{C}^{0|2}$ is the Berezin integral $\int d\varepsilon_1\, d\varepsilon_2$, which picks out the top component in the odd directions.

Construction 0.2.2 *(The open field from the Ext basis).* Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each a_I is a de Rham form on $\mathbb{R}$ with values in $\mathfrak{gl}_N$, shifted by $[1]$. The ghost-number-zero components are exactly

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N, \qquad \phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and they appear as

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The Berezin integral kills every monomial except the coefficient of ε_{12}, so every term in the ghost-zero action is obtained by multiplying the displayed three summands and keeping exactly one real one-form and exactly the product $\varepsilon_1\varepsilon_2$.

Remark 0.2.3. The shift [1] acts on the total cohomological degree: a form of de Rham degree p and Grassmann degree q has total degree $p+q$, which after the shift sits in ghost number $p+q-1$. In particular, the 1-form component of α (de Rham degree 1, Grassmann degree 0) sits in ghost number 0 and is the bosonic gauge field. The scalar components proportional to ε_i (de Rham degree 0, Grassmann degree 1) also sit in ghost number 0 and represent the motion of the brane in the two normal directions.

0.2.3.1 Ghost-zero open field content

The ghost-number-zero part of the BV field is

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2,$$

where $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$ and $\phi_i \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$.

LEMMA 0.2.4. With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to α_0 is, up to a boundary term on $\mathbb{R}$,

$$S(\alpha_0) = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Proof. Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive the integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In the quadratic term this leaves

$$\frac{1}{2}\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 - \phi_2\, d\phi_1\big),$$

where the sign is the Koszul sign from interchanging the two odd generators. Since $d\,\mathrm{Tr}(\phi_1\phi_2) = \mathrm{Tr}(d\phi_1\,\phi_2 + \phi_1\, d\phi_2)$, the second summand differs from $\mathrm{Tr}(\phi_1\, d\phi_2)$ by an exact one-form. In the cubic term the six orderings of $A, \phi_1\varepsilon_1, \phi_2\varepsilon_2$ reduce by cyclicity of the trace and the odd relation $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$ to $\mathrm{Tr}(A\phi_1\phi_2 - A\phi_2\phi_1)$. This is $\mathrm{Tr}(A[\phi_1, \phi_2])$. □

PROPOSITION 0.2.5 *(Classical BV truncation).* The subspace of ghost-number-zero fields

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$$

with all nonzero ghost-number components set to zero is a consistent classical truncation of the open BV theory. The induced classical action is the first-order gauged matrix model of Lemma 0.2.4.

Proof. Decompose the BV field as $\alpha = \sum_r \alpha^{(r)}$ by ghost number. The differential $d_{\mathbb{R}}$ and the product on $\Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N$ preserve total degree, and the BV action has ghost number zero. Hence every monomial in the Euler-Lagrange expression for a component of ghost number $r \neq 0$ contains at least one field component of nonzero ghost number when evaluated at $\alpha = \alpha_0$. Setting all such components to zero therefore makes the discarded equations vanish. The remaining equation is exactly the variation of $S(\alpha_0)$, computed in Lemma 0.2.4.

Concretely, the full Euler-Lagrange expression is the curvature

$$F(\alpha) = d\alpha + \alpha^2.$$

For $\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$ on a line,

$$F(\alpha_0) = (d\phi_1 + [A, \phi_1])\varepsilon_1 + (d\phi_2 + [A, \phi_2])\varepsilon_2 + [\phi_1, \phi_2]\varepsilon_1\varepsilon_2.$$

The BV pairing keeps exactly real-form degree one and $\varepsilon_1\varepsilon_2$. Hence the displayed three components pair exactly with the variations of A, ϕ_2, ϕ_1 respectively, while variations of nonzero ghost number see complementary curvature components that are zero for α_0 by real-form degree or ε-degree. This is a physical ghost-zero truncation; the residual ordinary GL_N gauge symmetry remains. □

The ghost-number-zero truncation of the BV master action is closed under the classical equations and gives a quantum-mechanical system with a connection form

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

two adjoint scalars

$$\phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and action

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

This action is invariant under

$$\delta_\varepsilon \phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon],$$

for $\varepsilon \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$. Indeed the variation of $\mathrm{Tr}(\phi_1 d\phi_2)$ containing $d\varepsilon$ is $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, and this cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing.

Remark 0.2.6. The same open model is obtained inside the Calabi-Yau category by tensoring a fixed object with $\mathbb{C}^N$ and passing to endomorphisms. The categorical construction is governed by cyclic cohomology. The three-point function is the cup product in cyclic cohomology followed by the trace; higher n-point functions require the corresponding higher operations.

Remark 0.2.7. In AKSZ language this is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ and symplectic potential $\mathrm{Tr}(\phi_1\, d\phi_2)$. We use the resulting Poisson bivector

$$\Pi = \sum_{i,j} \frac{\partial}{\partial(\phi_1)^i{}_j} \wedge \frac{\partial}{\partial(\phi_2)^j{}_i},$$

so $\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$. Equivalently, with the convention $\iota_{X_f}\omega = -df$ used throughout, the symplectic form is $\mathrm{Tr}(d\phi_1 \wedge d\phi_2)$. For $\epsilon \in \mathfrak{gl}_N$, the Hamiltonian

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big)$$

generates $\delta_\epsilon \phi_i = [\epsilon, \phi_i]$. Its zero locus is the commuting-pair equation $[\phi_1, \phi_2] = 0$, and the classical reduced phase space is the stack of commuting pairs of matrices modulo conjugation.

Remark 0.2.8. Equivalently, take Chern-Simons theory on $\mathbb{R} \times T^2$ and restrict to the harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on T^2. Writing $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$ and integrating over T^2 gives the same action; the odd variables ε_i represent the degree-one cohomology classes η_i.

0.2.4 *Closed String States*

Similarly, the closed string state space is of the form

$$\text{A-model closed states} \otimes \text{B-model closed states.}$$

For the A-model on $\mathbb{R}^2$, we take the de Rham cohomology complex $\Omega^*(\mathbb{R}^2)$.

For the B-model on $\mathbb{C}^2$, we use the local holomorphic-symplectic Kodaira-Spencer model obtained from Hamiltonian holomorphic vector fields. It uses polyvectors and the divergence operator, and it is distinct from a truncation of Calabi-Yau threefold BCOV theory.

Remark 0.2.9. The original BCOV analysis [1] is written for Calabi-Yau threefolds. On a holomorphic symplectic surface the natural local cotangent theory is the BF theory for the dg Lie algebra of Hamiltonian vector fields. The holomorphic symplectic form supplies the Poisson bracket, while the holomorphic volume form supplies the divergence operator on polyvectors.

We consider the space of polyvector fields

$$\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$$

where ∂_{z_i} are odd generators of poly-vector degree 1 (corresponding to the two holomorphic vector fields on $\mathbb{C}^2$). The operator

$$\partial : \mathbf{PV}^{i,*} \to \mathbf{PV}^{i-1,*}$$

is the divergence operator determined by the holomorphic volume form $\Omega = dz_1 \wedge dz_2$:

$$\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\mathrm{hol}} \circ \iota_\Omega, \qquad \iota_\Omega : \mathbf{PV}^{i,*} \xrightarrow{\sim} \Omega^{2-i,*}.$$

Thus for $X = a_1\partial_{z_1} + a_2\partial_{z_2}$,

$$\partial_\Omega X = (\partial_{z_1} a_1 + \partial_{z_2} a_2),$$

while ∂_Ω vanishes on functions.

We work in the formal polynomial local model at $0 \in \mathbb{C}^2$, so Dolbeault cohomology is represented by holomorphic Taylor polynomials. Here ∂_Ω is the divergence constraint selecting the Hamiltonian BF sector. The kinematic differential of the closed theory below is $D = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$. Thus the Poincare lemma identifies divergence-free vector fields with Hamiltonian potentials. The quotient by ∂_Ω-exact Hamiltonians is absent. Analysing degree by degree:

- $\mathbf{PV}^{0,*}(\mathbb{C}^2) = \Omega^{0,*}(\mathbb{C}^2) \subset \mathbf{Ker}\partial$ trivially.
- $\mathbf{PV}^{1,*}(\mathbb{C}^2) \cong \Omega^{1,*}(\mathbb{C}^2)$ via $\iota_{(-)}\omega$. Under this identification, with the Hamiltonian convention $\iota_{X_f}\omega = -df$ used below, $\mathbf{Ker}\partial_\Omega$ is the space of closed holomorphic one-forms. Lemma 0.2.10 identifies these with df, uniquely modulo constants.
- The constant Hamiltonians act trivially, since their Hamiltonian vector fields vanish. We therefore use the reduced polynomial algebra
$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$$
for nontrivial Hamiltonian fields.

LEMMA 0.2.10 *(Polynomial Poincare primitive).* Let

$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$

be a polynomial one-form on $\mathbb{C}^2$. If $\partial_{z_2} a = \partial_{z_1} b$, then there is a polynomial $f \in \mathbb{C}[z_1, z_2]$, unique modulo constants, such that $\theta = df$.

Proof. Define

$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$

Since a and b are polynomials, the integral is termwise division of each homogeneous term by its degree plus one, hence f is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so

$$\begin{aligned}
\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\
&= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\
&= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).
\end{aligned}$$

The same calculation, with a and b interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. □

Proposition 0.2.11 *(Hamiltonian polyvector reduction).* In the formal polynomial model on $(\mathbb{C}^2, \omega = dz_1 \wedge dz_2)$, the divergence-closed polyvector-one fields reduce to Hamiltonian vector fields:

$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$

Under the convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df,$$

the commutator of vector fields is

$$[X_f, X_g] = X_{\{f,g\}_{\bar{A}}}.$$

Proof. Let $X = a_1 \partial_{z_1} + a_2 \partial_{z_2}$. The divergence operator is $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. Contracting with ω gives

$$\iota_X \omega = a_1\, dz_2 - a_2\, dz_1.$$

The equation $\partial_\Omega X = 0$ is exactly the closedness of this one-form:

$$d(\iota_X \omega) = (\partial_{z_1} a_1 + \partial_{z_2} a_2)\, dz_1 \wedge dz_2 = 0.$$

By Lemma 0.2.10, there is a polynomial f, unique modulo constants, such that $\iota_X \omega = -df$. Therefore

$$a_1 = -\partial_{z_2} f, \qquad a_2 = \partial_{z_1} f,$$

so $X = X_f$. Constants are exactly the kernel of $f \mapsto X_f$, giving $\bar{A}$.

Finally, for any polynomial h,

$$\begin{aligned}
[X_f, X_g](h) &= \{f, \{g, h\}\} - \{g, \{f, h\}\} \\
&= \{\{f, g\}, h\}
\end{aligned}$$

by the Jacobi identity for the constant Poisson bracket. Thus $[X_f, X_g] = X_{\{f,g\}}$, and projecting constants gives the bracket on $\bar{A}$. □

Problem 0.2.12 *(Analytic graph realization data).* A closed-string Feynman-integral theorem on the noncompact space $\mathbb{R}^2 \times \mathbb{C}^2$ requires the following data:

(i) a global BV elliptic complex extending the formal jet model;
(ii) boundary or asymptotic conditions on $\mathbb{C}^2$;
(iii) a gauge fixing Q^{GF} and the corresponding heat-kernel propagator

$$P_{\epsilon,L} = \int_{\epsilon}^{L} Q^{\mathrm{GF}} K_t \, dt;$$

(iv) the treatment of zero modes and the infrared limit $L \to \infty$;
(v) the ultraviolet renormalisation scheme and local counterterms;
(vi) a proof of the quantum master equation, or the exact anomaly class;
(vii) a bulk-to-boundary kernel for the brane;
(viii) trace normalisation and large-N weighting conventions;
(ix) a convention for extracting connected single-trace and multi-trace coefficients.

The computations in these notes use the formal local algebra of polynomial Hamiltonians. They are independent of the analytic choices above.

For the Hamiltonian sector of the closed string field theory we replace $\mathbf{Ker}\partial$ (on poly-1-vectors) by the Dolbeault-valued Hamiltonian vector fields:

$$\Omega^{0,*}(\mathbb{C}^2)\text{-valued Hamiltonian vector fields,}$$

i.e. sections of $\Omega^{0,*}(\mathbb{C}^2)$ identified with Hamiltonian vector fields via the symplectic form $\omega = dz_1 \wedge dz_2$. Concretely, a section $f \in \Omega^{0,*}(\mathbb{C}^2)$ corresponds to the Hamiltonian vector field $X_f = \{f, -\}_\omega$, where

$$\{f, g\}_\omega = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g.$$

With this convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df.$$

Remark 0.2.13. The replacement is the elementary statement of Proposition 0.2.11. Constants are removed because $X_f = 0$ exactly when f is constant.

0.2.5 Closed String Field Theory

0.2.5.1 The Action Functional

Let

$$D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

Write

$$\alpha \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \otimes \bar{A}[1]$$

for the Hamiltonian connection field and

$$\beta \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \otimes \bar{A}^\vee[2]$$

for its cotangent field. The bracket on form-valued Hamiltonians is obtained by wedging the form parts and taking the Poisson bracket of the Hamiltonian parts with the tensor-product Koszul sign:

$$[\eta \otimes f, \theta \otimes g] = (-1)^{|f||\theta|}(\eta \wedge \theta) \otimes \{f, g\},$$

for homogeneous form η, θ and homogeneous Hamiltonian coefficients f, g. In the ghost-zero Hamiltonian calculations used below the coefficients are even, so this sign is +1. The action is

$$S = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \left\langle \beta,\ D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle dz_1\, dz_2.$$

The pairing is the evaluation pairing between $\bar{A}^\vee$ and $\bar{A}$ (or, in a formal residue model, the corresponding continuous residue pairing). For ghost-number-zero fields, α has total real/Dolbeault form degree 1 and β has total degree 2. Hence $\beta \wedge D\alpha$ and $\beta \wedge \{\alpha, \alpha\}$ have total real/Dolbeault degree 4; multiplying by $dz_1 \wedge dz_2$ gives a top real degree 6 form on $\mathbb{R}^2 \times \mathbb{C}^2$.

Remark 0.2.14. The open quantum-mechanical system lives on the brane line. The closed Hamiltonian BF theory lives on the full space-time.

0.2.5.2 Classical field content

This theory is closely related to a 4-dimensional BF theory: β plays the role of the B-field (Lagrange multiplier) and the equations of motion of β enforce $F_\alpha = 0$, where

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha, \alpha\}$$

is the "curvature" of α (viewed as a connection on a bundle of holomorphic-symplectic surfaces over $\mathbb{R}^2$).

Consider the space-time $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$. The ghost-number-zero components of α and β are:

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar{z}_j}\, d\bar{z}_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1\, dx_2 + \beta_{x_i \bar{z}_j}\, dx_i\, d\bar{z}_j + \beta_{\bar{z}_1 \bar{z}_2}\, d\bar{z}_1\, d\bar{z}_2.$$

Here the coefficients of α are Hamiltonian functions modulo constants. The gauge symmetries are the BF-type transformations

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \mathrm{ad}^\vee_\varepsilon\, \beta,$$

and

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

where $D_\alpha = D + \{\alpha, -\}$ and λ is a one-form valued in $\bar{A}^\vee$. The second symmetry is reducible on shell: if $\lambda = D_\alpha \mu$, then $D_\alpha \lambda = D_\alpha^2 \mu = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Analysing α as a Hamiltonian function on $\mathbb{C}^2$: the component $\alpha_{\bar{z}_j} d\bar{z}_j$ is the Hamiltonian potential whose vector field $X_{\alpha_{\bar{z}_j}} d\bar{z}_j$ is the Beltrami differential encoding a deformation of the complex structure of $\mathbb{C}^2$; the component $\alpha_{x_i} dx_i$ similarly defines a connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields on $\mathbb{C}^2$, encoding a bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. Together, the equation of motion $F_\alpha = 0$ says this data constitutes an integrable deformation of complex structure with flat holonomy. The field β appears linearly in the action and enforces exactly this equation.

Remark 0.2.15. This is entirely analogous to BF theory, where B is a Lagrange multiplier enforcing $F_A = 0$, and the phase space is $T^*\mathcal{M}_{\mathrm{flat}}$.

0.2.5.3 Phase Space

Formally compactify one real direction and put the perturbative local model on $\mathbb{R} \times S^1 \times X$, where X is holomorphic symplectic and the calculation is completed at the trivial solution. The equations of motion $F_\alpha = 0$ cut out the Maurer-Cartan moduli problem for Hamiltonian vector fields on $S^1 \times X$.

Fix the analytic package of Problem 0.2.12 in this compact or formal setting: elliptic complex, boundary condition, gauge fixing, propagator, renormalization scheme, and anomaly cancellation. The perturbative phase space is the derived cotangent stack

$$T^*_{\mathrm{der}}\mathrm{MC}\big(\Omega^*(S^1)\widehat{\otimes}\Omega^{0,*}(X,\mathrm{Ham}(X)),D,\{-,-\}\big).$$

After passing to the underived tangent space at the trivial solution this reduces to the ordinary cotangent bundle of the linearized deformation complex.

0.2.6 *Closed String Fields and Brane Deformations*

The local classical bulk-boundary operation is evaluation of a closed Hamiltonian on the brane normal fields. If the brane lies along x_1 and the closed string field is $dx_1\, z_1^k z_2^l$, the local boundary functional is

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1,\phi_2]\big) + \int_{\mathbb{R}} dx_1\, \mathrm{Tr}\, \phi_1^k \phi_2^l.$$

This is a statement inside the formal local algebraic model: the closed Hamiltonian is pulled back along the matrix-valued normal coordinate map $z_i \mapsto \phi_i$ and then traced. Construction 0.2.21 is the algebraic form of this bulk-to-boundary prescription, and Theorem 0.2.40 below proves that the resulting Hamiltonian evaluation preserves the classical bracket.

Remark 0.2.16. Geometrically, specifying a closed string field on $\mathbb{C}^2$ amounts to equipping $\mathbb{C}^2$ with a connection, so that the brane can parallel-transport its internal degrees of freedom as it moves along $\mathbb{R}$.

0.2.7 *Operators in the Gauge Theory and the Large N Limit*

The local-operator calculation is performed in the reduced BV model with fields ϕ_1, ϕ_2 and the antifield $\psi = A^*$ for the Lagrange multiplier A. We take GL_N-invariants in characteristic zero, so the Chevalley part of the gauge BRST differential acts trivially on trace functions by cyclicity.

Construction 0.2.17 *(The reduced BV algebra at the insertion point).* At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its BRST differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1,\phi_2])^i{}_j.$$

This formula is the algebraic shadow of the A-equation of motion: varying

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1,\phi_2]\big)$$

with respect to A gives $[\phi_1,\phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

Lemma 0.2.18 *(De Rham jets).* Positive jets in the topological direction vanish in local operator cohomology. Equivalently, every cohomology class has a representative determined by the values of the fields and antifields at the insertion point.

Proof. For a coordinate t on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive t-derivative is cohomologous to zero after applying the usual Euler homotopy on jet degree. This kills topological-direction jets and leaves the holomorphic z_i-derivatives used later on the closed side. □

Proposition 0.2.19 *(Stable trace invariants).* Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F,F]$ be the vector space of cyclic words. For each N, GL_N acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The degreewise stable invariant algebra

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} := \bigoplus_{d\geq 0} \varprojlim_N \big(\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}\big)_d$$

with respect to block-restriction maps in each word-degree is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word w maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$. More precisely, in total positive word-degree d this map is an isomorphism for every $N \geq d$, after removing the scalar empty trace at finite N.

Proof. Procesi–Razmyslov invariant theory [6]razmyslov gives two inputs. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size N; in particular every homogeneous trace relation has total word-degree at least $N+1$. Fix a total word-degree d. For every $N \geq d$ the degree-d relations are exactly cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-N trace identities. The surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words.

For the one- and two-ψ sectors used below, start with a third even generic matrix, take the part linear or quadratic in that third letter, and then skew-symmetrise the marked letters. This is the same as declaring the marked letter ψ odd. The stable single-trace relation is graded cyclicity: moving one ψ through the trace has sign $+1$, while moving one ψ past another gives a minus sign. These are exactly the cyclic signs used in the finite complex below. □

Remark 0.2.20. The empty trace is the finite-rank scalar $\mathrm{Tr}_N(1) = N$, which changes with N and fails to define a stable primitive generator. We remove this scalar line before passing to the degreewise stable limit. The class $\mathrm{Tr}(\psi)$ is a genuine stable odd primitive class. It is excluded from the Hamiltonian comparison because constants have been removed from the closed Hamiltonian algebra.

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

Construction 0.2.21 *(Boundary evaluation of a closed Hamiltonian).* Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal coordinates z_1, z_2 by the two adjoint scalar fields ϕ_1, ϕ_2. Choosing the ordered lift with all ϕ_1's to the left of all ϕ_2's gives the invariant function

$$\widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \operatorname{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Let

$$H^0_{\mathrm{red}}(\mathcal{R}_N^{GL_N}, Q) := H^0(\mathcal{R}_N^{GL_N}, Q)/\mathbb{C} \cdot \operatorname{Tr}(1).$$

Constants give the scalar $c_{0,0}N$ and therefore vanish in this quotient. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0_{\mathrm{red}}(\mathcal{R}_N^{GL_N}, Q), \qquad [f] \longmapsto \overline{\widetilde{\partial}_{\mathrm{bb},N}(f)}.$$

Let $\mathsf{W}_{k,l}$ be the set of words in two letters containing k copies of the first letter and l copies of the second. The first descendant is obtained by inserting one Koszul variable and summing over all ordered lifts: For $k + l > 0$ the descendant part of the Hamiltonian comparison is

$$\partial^{(1)}_{\mathrm{bb},N}(\delta_{k,l}[1]) = \Psi_{k,l}, \qquad \Psi_{k,l} := \sum_{w \in \mathsf{W}_{k,l}} \operatorname{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

The same formula at $(k, l) = (0, 0)$ gives the decoupled scalar antifield $\operatorname{Tr}(\psi)$, which is excluded from the Hamiltonian pairing because constants have been removed from $\bar{A}$.

LEMMA 0.2.22 *(First descendants are cycles).* For every $k, l \geq 0$, the symmetrised trace $\Psi_{k,l}$ satisfies $Q\Psi_{k,l} = 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Since x and y are even and $Q\psi = [x, y]$, one has

$$Q\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \operatorname{Tr}\big((xy - yx)w(x, y)\big).$$

Group the resulting traces by cyclic word of multidegree $(k + 1, l + 1)$. The coefficient of a cyclic word in the positive sum is the number of cyclic adjacent occurrences of the block xy; the coefficient in the negative sum is the number of cyclic adjacent occurrences of the block yx. On a circle the number of transitions from an x-run to a y-run equals the number of transitions from a y-run to an x-run. The two grouped sums are therefore equal, so $Q\Psi_{k,l} = 0$. □

PROPOSITION 0.2.23 *(Primitive one-antifield Koszul homology).* Fix $k, l \geq 0$. The middle homology of the primitive single-trace complex with one displayed ψ and k, l displayed x, y letters is one-dimensional, generated by the class of $\Psi_{k,l}$. The nonconstant part relevant to the Hamiltonian comparison is the direct sum over $k + l > 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. Primitive cyclic words are taken modulo the graded cyclic rule

$$\operatorname{Tr}(ab) = (-1)^{|a||b|}\operatorname{Tr}(ba), \qquad |x| = |y| = 0, \quad |\psi| = 1.$$

Graded cyclicity gives a unique representative $\operatorname{Tr}(\psi w)$ for every one-ψ cyclic word. Thus the relevant one-ψ chain group is

$$C_1(k, l) = \mathbb{C}\{\mathsf{W}_{k,l}\}.$$

The zero-ψ target is

$$C_0(k,l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and the differential is the incidence map

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-ψ source is generated by traces $\mathrm{Tr}(\psi u \psi v)$ with u and v ordinary words whose total bidegree is $(k-1, l-1)$; if $k = 0$ or $l = 0$ this source is zero. Since ψ is odd, exchanging the two displayed ψ's changes sign. Applying $Q\psi = xy - yx$ gives

$$\begin{aligned}\partial_2(u,v) &= Q\,\mathrm{Tr}(\psi u \psi v)\\ &= \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big),\end{aligned}$$

so, after removing the common front ψ,

$$\partial_2(u,v) = vxyu - vyxu - uxyv + uyxv.$$

A direct substitution shows $\partial_1\partial_2 = 0$.

If $k = 0$ or $l = 0$, then $C_1(k,l)$ is spanned by x^k or y^l, $\partial_1 = 0$, and $C_2(k,l) = 0$. Hence the middle homology is already the line spanned by $\Psi_{k,l}$. Assume from now on that $k, l > 0$.

We compute the first homology of this finite complex. Form a graph $\Gamma_{k,l}$ whose vertices are the necklaces in $\mathsf{N}_{k+1,l+1}$ and whose oriented edges are the words $w \in \mathsf{W}_{k,l}$, with

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Then ∂_1 is the cellular boundary of this graph. For each pair (u,v) as above attach the square whose oriented boundary is $vxyu - vyxu - uxyv + uyxv$. Denote the resulting two-complex by $X_{k,l}$.

Unwrap a necklace at a marked y. A vertex is then a tuple

$$(a_0, \dots, a_l), \qquad a_i \geq 0, \qquad \sum_i a_i = k+1,$$

where a_i counts the number of x's between the ith and $(i+1)$st y. An edge moves one x across one neighbouring y; in these coordinates it replaces a_i, a_{i+1} by $a_i - 1, a_{i+1} + 1$ when $a_i > 0$. The attached squares say that two moves of two chosen x's commute whenever both orders of the moves are defined.

Lift the cyclic set of y-gaps from $\mathbb{Z}/(l+1)$ to $\mathbb{Z}$. The lifted complex has vertices

$$p_0 \leq \cdots \leq p_k, \qquad p_k - p_0 \leq l,$$

where p_i records the lifted gap occupied by the ith x. Edges change one p_i by one, preserving the displayed inequalities, and squares commute changes of two different coordinates. This is the cubical subdivision of the convex polyhedron

$$\{(t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_0 \leq \cdots \leq t_k,\ t_k - t_0 \leq l\},$$

hence its first homology vanishes. The original necklace complex is the quotient by the free deck transformation

$$(p_0, \dots, p_k) \longmapsto (p_0 + l + 1, \dots, p_k + l + 1),$$

which moves all x's once around the circle of y-gaps. Therefore $H_1(X_{k,l};\mathbb{C})$ is one-dimensional. The cellular one-cycle $\sum_{w\in \mathsf{W}_{k,l}} w$ lifts to a path from a vertex of the lifted complex to its deck translate, so it is the generator of this deck-circle homology.

Lemma 0.2.22 identifies this representative as a cycle in the original trace complex. Its non-boundary property follows from the functional

$$\epsilon\big(\mathrm{Tr}(\psi w)\big) = 1, \qquad w \in \mathsf{W}_{k,l},$$

kills every displayed two-ψ boundary because each boundary has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Thus the one-ψ homology in bidegree (k,l) is the one-dimensional line generated by $[\Psi_{k,l}]$. □

Remark 0.2.24. The script `scripts/check_one_psi_homology.py` constructs the finite complex $C_2(k,l) \to C_1(k,l) \to C_0(k,l)$ over $\mathbb{Q}$ and verifies $\mathbf{dim}(\ker)\partial_1/\mathrm{im}\,\partial_2 = 1$ in low bidegrees. It also checks the open Hamiltonian action of Proposition 0.2.38 in low bidegrees. These checks verify the displayed finite models without adding assumptions.

COROLLARY 0.2.25 *(Boundary pairing for cotangent generators).* Let $\delta_{k,l} \in \mathfrak{h}^\vee$ be the functional dual to the monomial $z_1^k z_2^l \in \bar{A}$. The assignment

$$\delta_{k,l}[1] \longmapsto [\Psi_{k,l}]$$

extends to an isomorphism from the shifted Taylor-dual space $\mathfrak{h}^\vee[1]$ to the nonconstant primitive one-ψ brane homology.

Proof. The monomials $z_1^k z_2^l$ with $k+l>0$ form a topological basis of $\bar{A}$, and the $\delta_{k,l}$ form the continuous dual basis. By Proposition 0.2.23, the primitive one-ψ homology has one generator in each bidegree, namely $[\Psi_{k,l}]$. Discarding the scalar bidegree $(0,0)$ and sending the dual basis vector in every nonconstant bidegree to that generator gives a filtered vector-space isomorphism. This is the boundary pairing used in the PBW special fibre; the closed module structure remains coadjoint. □

LEMMA 0.2.26 *(Adjacent swaps).* In the invariant reduced BV cohomology,

$$\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v)$$

is Q-exact for all words u, v.

Proof. By cyclicity of the trace,

$$\begin{aligned} Q\,\mathrm{Tr}(\psi vu) &= \mathrm{Tr}([\phi_1,\phi_2]vu) \\ &= \mathrm{Tr}(\phi_1\phi_2 vu - \phi_2\phi_1 vu) \\ &= \mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Corollary 0.2.27 *(Matrix evaluation is commutative in cohomology).* If two words in ϕ_1, ϕ_2 have the same numbers of ϕ_1 and ϕ_2, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ is determined by the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$ and is independent of the chosen ordered lift.

Proof. Adjacent transpositions generate all reorderings of a finite word. Each adjacent transposition of the adjacent letters ϕ_1, ϕ_2 changes the trace by a Q-exact term by Lemma 0.2.26; cyclicity moves the chosen adjacent pair into the displayed position of that lemma. Summing the resulting exact differences proves the claim. □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\mathrm{Tr}(1) = N$ and has zero Hamiltonian vector field.

Proposition 0.2.28 *(Brane Hamiltonian sector).* In the stable large-N scalar-reduced BV model, the Hamiltonian descendant algebra freely generated by the nonconstant primitive ghost-zero traces and the nonconstant primitive one-ψ Koszul homology is

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big).$$

The class of $z_1^k z_2^l$ with $k + l > 0$ corresponds to $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$. The corresponding descendant basis vector corresponds to the symmetrised trace with one ψ and k, l copies of ϕ_1, ϕ_2.

Proof. Proposition 0.2.19 gives the free commutative algebra on cyclic words. Lemma 0.2.26 identifies any two cyclic words with the same numbers of ϕ_1 and ϕ_2, because adjacent transpositions generate all reorderings. Thus the scalar-reduced ghost-zero cyclic words reduce to the monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$. Proposition 0.2.23 computes the primitive one-ψ homology in each multidegree; the nonconstant generators are $\Psi_{k,l}$ with $k + l > 0$. Products of the scalar-reduced ghost-zero primitive classes and these one-ψ primitive classes give the free graded-commutative multi-trace algebra in the Hamiltonian sector. By definition of the Hamiltonian sector, single traces containing two or more occurrences of ψ are excluded from this algebra. □

Remark 0.2.29. The word "sector" is literal. The calculation above presents the stable algebra generated by the primitive no-ψ Hamiltonians and the nonconstant primitive one-ψ homology generated by $\Psi_{k,l}$ for $k + l > 0$. The full invariant Koszul homology in higher ψ-count lies outside this sector, as does the decoupled scalar class $\mathrm{Tr}(\psi)$.

Problem 0.2.30 *(Full primitive Koszul homology).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| = 1,$$

with differential $d\psi = xy - yx$ and $dx = dy = 0$. Put

$$\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give x, y, ψ weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed (R, S) define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R, S)$. The differential is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} x y\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} y x\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

Prove the primitive homology formula

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \ge 1, \\ 0, & p \ge 2. \end{cases}$$

Equivalently, prove that the primitive stable Koszul homology is $\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta$ with $|\theta| = 1$ and $\mathrm{wt}(\theta) = (1, 1)$.

The required new ingredient is the all-p orientation statement identifying this graded cyclic complex with the cellular chains of the cubical necklace complex whose universal cover is contractible and whose deck quotient has homotopy type S^1. Proposition 0.2.23 proves exactly the $p \le 2$ skeleton needed for the Hamiltonian sector. The complete theorem also requires the all-ψ stable trace presentation and an exact computational check of the finite complex.

0.2.8 Operators in the Gravitational Theory

Let $\mathfrak{h} = \widehat{\mathfrak{h}} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ with Poisson bracket, and let $\mathfrak{h}^\vee$ be its continuous Taylor dual. The cotangent field is a coadjoint module, so the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

For $f, g \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^\vee$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0,$$

where $(\mathrm{ad}_f^\vee \lambda)(g) = -\lambda(\{f, g\})$. On polynomial monomials in the dense subalgebra $\bar{A}$ this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

where the bar means projection to $\bar{A}$; in particular the constant bracket of z_1 and z_2 is zero in the reduced Hamiltonian Lie algebra.

LEMMA 0.2.31 *(Shifted coadjoint signs).* Let η_λ denote the suspension of $\lambda \in \mathfrak{h}^\vee$ into $\mathfrak{h}^\vee[1]$. Since the Hamiltonians in $\mathfrak{h}$ are even,

$$[f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Consequently the enveloping algebra relations involving two shifted coadjoint generators are odd-commutative:

$$\eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda = 0.$$

Proof. The semidirect product bracket before shifting is the ordinary coadjoint action of the even Lie algebra $\mathfrak{h}$ on its continuous dual. Suspending the module raises the degree of every λ by one. The mixed-bracket sign is $+1$ because $|f| = 0$. The shifted module is abelian as a Lie ideal, so its bracket with itself is zero; in the associative enveloping algebra this zero Lie bracket gives $\eta_\lambda \eta_\mu - (-1)^{1\cdot 1} \eta_\mu \eta_\lambda = 0$. □

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 0.2.18, together with the Dolbeault Poincare lemma in the formal local model, leaves precisely the holomorphic Taylor coefficients. Thus the linear local coordinates are the functionals

$$\partial_{z_1}^k \partial_{z_2}^l \alpha|_0, \qquad \partial_{z_1}^k \partial_{z_2}^l \beta|_0.$$

These are dual to the Taylor monomials $z_1^k z_2^l$ up to the factorial normalisation fixed by the chosen pairing. Since zero-form components of α and β have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

Proposition 0.2.32 *(Closed local operators are continuous CE cochains).* The classical local operators of the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} \cong C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}^{\vee}_{\mathrm{cont}}[-1]),$$

with CE differential dual to the Lie brackets above.

Proof. Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of α and β. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The Q_0-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining α-coefficients are Taylor Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \widehat{\mathfrak{h}}$; degreewise polynomial coefficients form the dense subalgebra $\bar{A}$. The cotangent field β supplies the odd shifted continuous dual $\mathfrak{h}^{\vee}[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}[1]$.

The nonlinear BRST differential is determined by the BV Hamiltonian

$$S = \int \left\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle.$$

The linear part is exactly Q_0, already killed in cohomology. The quadratic term sends two Hamiltonian α-modes f, g to their reduced Poisson bracket $\{f, g\}$, and the action on the cotangent variable is the coadjoint action displayed above. The quadratic term has zero component with two cotangent inputs, so $[\eta_\lambda, \eta_\mu] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

0.2.9 Koszul Duality

Definition 0.2.33. For an augmented complete dg algebra B, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C}).$$

We regard commutative Chevalley-Eilenberg cochains as E_1 algebras by restriction. All bar and cobar constructions below are taken in the category of complete filtered Tate vector spaces, with continuous maps. The completions are by symmetric or tensor length together with the coefficient topology coming from Taylor degree.

LEMMA 0.2.34 *(Tate continuous bar-cobar).* Let $V = \prod_{d\geq 1} V_d$ and $W = \bigoplus_{d\geq 1} W_d$, with all V_d, W_d finite dimensional. Then

$$(V \oplus W)^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} V_d^\vee \oplus \prod_{d\geq 1} W_d^\vee.$$

Let $\mathfrak{g}$ be a Tate graded Lie algebra whose bracket is continuous for such a coefficient topology and whose finite coefficient windows are finite dimensional. Then

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \mathrm{Hom}_{\mathrm{cont}}\big(\mathbf{S}^c(\mathfrak{g}[1]), \mathbb{C}\big) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

The continuous bar resolution of the augmentation has dual cobar algebra $\Omega C_*^{\mathrm{CE}}(\mathfrak{g})$, and the canonical Lie bar-cobar map

$$\Omega C_*^{\mathrm{CE}}(\mathfrak{g}) \longrightarrow \widehat{U}(\mathfrak{g})$$

is a filtered quasi-isomorphism. Its associated graded map is the completed PBW isomorphism

$$\mathrm{gr}\,\widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}(\mathfrak{g}).$$

Proof. A continuous functional on $\prod V_d$ depends on finitely many homogeneous coordinates. A continuous functional on the discrete direct sum $\bigoplus W_d$ is an arbitrary compatible product of degreewise functionals. This proves the displayed duality formula.

Filter the CE coalgebra by symmetric coalgebra length and by finite coefficient windows. After projecting to a fixed output window and bounding tensor length, continuity of the bracket says that finitely many homogeneous input coefficients can contribute. The corresponding finite diagram is the ordinary finite-dimensional Lie bar-cobar comparison, where the bar resolution computes $\mathrm{RHom}_{C^\bullet_{\mathrm{CE}}}(\mathbb{C}, \mathbb{C})$ and the cobar algebra maps to the universal enveloping algebra. These finite comparisons are compatible under enlargement of the window. Passing to the Tate limit and completing in tensor length gives the asserted continuous cobar algebra. The PBW filtration by tensor length has associated graded the completed symmetric algebra, so the completed comparison is a filtered quasi-isomorphism. □

PROPOSITION 0.2.35 *(Koszul dual of continuous CE cochains).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ be the completed graded Lie algebra above, and let

$$B = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

Then

$$B^! \simeq \widehat{U(\mathfrak{g})},$$

the completed graded universal enveloping algebra.

Proof. Put

$$H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\}, \qquad \mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee.$$

Thus

$$\mathfrak{g} = \left(\prod_{d\geq 1} H_d\right) \ltimes \left(\bigoplus_{d\geq 1} H_d^\vee\right)[1]$$

is a Tate Lie algebra. The bracket is continuous: the degree-r coefficient of $\{f, g\}$ is determined by Taylor coefficients of f and g of degrees at most $r + 1$, since ∂_{z_1} and ∂_{z_2} each lower degree by one. The

coadjoint action is continuous by the same finite-degree test. This Taylor topology is independent of the PBW/order filtration used for the Rees algebra below.

Let

$$C_*^{\mathrm{CE}}(\mathfrak{g}) = \mathbf{S}^c(\mathfrak{g}[1])$$

be the Chevalley-Eilenberg coalgebra, with coderivation determined by the bracket of $\mathfrak{g}$. By definition

$$B = \mathrm{Hom}_{\mathrm{cont}}\big(C_*^{\mathrm{CE}}(\mathfrak{g}), \mathbb{C}\big).$$

The coaugmentation filtration on $C_*^{\mathrm{CE}}(\mathfrak{g})$ is conilpotent in each finite Taylor window, and its completed Tate limit is pronilpotent. The augmentation $B \to \mathbb{C}$ has the continuous bar resolution, and Lemma 0.2.34 identifies

$$\mathrm{RHom}_B(\mathbb{C}, \mathbb{C}) \simeq \Omega\, C_*^{\mathrm{CE}}(\mathfrak{g}),$$

the completed cobar algebra of the CE coalgebra. The completed Lie bar-cobar comparison in the lemma identifies this cobar algebra with the completed associative algebra generated by $\mathfrak{g}$ modulo the closed graded Lie relations; this is $\widehat{U(\mathfrak{g})}$.

Thus, for the generators $f, g \in \mathfrak{h}$ and $\eta_\lambda, \eta_\mu \in \mathfrak{h}^\vee[1]$, the completed enveloping algebra has relations

$$\begin{aligned} fg - gf &= \{f, g\}, \\ f\eta_\lambda - \eta_\lambda f &= \eta_{\mathrm{ad}_f^\vee \lambda}, \\ \eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda &= 0. \end{aligned}$$

□

For comparison with the classical brane algebra introduce the completed PBW/order Rees enveloping algebra

$$U_\hbar(\mathfrak{g}) = \widehat{T}(\mathfrak{g})[[\hbar]] \Big/ \overline{\big\langle xy - (-1)^{|x||y|} yx - \hbar[x, y] \big\rangle},$$

where the closure is for tensor length and the Tate coefficient topology. This Rees parameter records word length in the enveloping algebra and is independent of Taylor degree. At $\hbar = 0$ one obtains the completed PBW special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

After choosing the monomial-dual Taylor basis $\mathfrak{h}^\vee \cong \bar{A}_{\mathrm{desc}}$, the degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$ is the stable classical brane Hamiltonian descendant algebra of Proposition 0.2.28. Thus the free, abelian closed special fibre is Koszul dual directly to this symmetric product algebra. The interacting closed Hamiltonian bracket deforms the closed special fibre to $\widehat{U(\mathfrak{g})}$; on the brane side the deformation checked in these notes is the degree-zero Hamiltonian trace bracket below.

0.2.9.1 First-order corrections, order hbar one

Under the Koszul duality correspondence we have

$$z_1^k z_2^l \leftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The reduced Poisson bracket on the gravity side is:

$$\{z_1^k z_2^l,\ z_1^m z_2^n\}_{\bar{A}} = \overline{(kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}}.$$

If an exponent in the displayed monomial is negative then the coefficient $kn - lm$ is already zero; if the displayed monomial is constant, its class in $\bar{A}$ is zero. The matching bracket on the open side is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

LEMMA 0.2.36 *(Hamiltonian reduction of the trace bracket).* The canonical matrix Poisson bracket descends to the scalar-reduced invariant quotient used above.

Proof. For $\epsilon \in \mathfrak{gl}_N$, the component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ of the moment map generates the infinitesimal conjugation action, with the sign fixed by the convention for the bracket. Hence $\{F, \mu_\epsilon\} = 0$ for every invariant function F. If I_μ is the ideal generated by all μ_ϵ, then $\{F, \mu_\epsilon K\} = \mu_\epsilon\{F, K\}$ lies again in I_μ. Therefore the bracket of invariant functions descends to $(\mathcal{R}_N/I_\mu)^{GL_N}$, equivalently to the degree-zero reduced BV cohomology. The scalar line $\mathbb{C} \cdot \mathrm{Tr}(1)$ is central, so quotienting by it is also compatible with the bracket. □

PROPOSITION 0.2.37 *(First-order trace bracket).* In the stable scalar-reduced cohomology,

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

Proof. Write $F_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial \phi_1}\frac{\partial G}{\partial \phi_2} - \frac{\partial F}{\partial \phi_2}\frac{\partial G}{\partial \phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial \phi_1} = \sum_{a=0}^{k-1} \phi_1^{k-1-a}\phi_2^l\phi_1^a,$$
$$\frac{\partial F_{k,l}}{\partial \phi_2} = \sum_{b=0}^{l-1} \phi_2^{l-1-b}\phi_1^k\phi_2^b.$$

Therefore the first summand in the bracket is a sum of kn traces, each obtained by removing one ϕ_1 from the first word and one ϕ_2 from the second word. The second summand is a sum of lm traces obtained by removing one ϕ_2 from the first word and one ϕ_1 from the second. Every trace appearing in either nonzero sum has multidegree $(k + m - 1, l + n - 1)$. Lemma 0.2.26 identifies all cyclic words with this multidegree in reduced BRST cohomology. The coefficient is therefore $kn - lm$, and the common class is $\overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}$. When the common word has multidegree $(0, 0)$, this class is the scalar trace and is zero in the scalar-reduced quotient. □

PROPOSITION 0.2.38 *(Open action on primitive descendants).* Let

$$\widetilde{\Psi}_{a,b} = \binom{a+b}{a}^{-1}\Psi_{a,b}$$

be the averaged primitive one-ψ class. In the stable primitive one-ψ homology,

$$\{\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

with the convention that a term with a negative exponent is zero. Thus the polynomial descendant classes carry the adjoint polynomial Hamiltonian action.

Proof. Put $x = \phi_1$ and $y = \phi_2$. The cyclic derivatives of $F_{p,q} = \mathrm{Tr}(x^p y^q)$ are

$$\frac{\partial F_{p,q}}{\partial x} = \sum_{i=0}^{p-1} x^{p-1-i} y^q x^i, \qquad \frac{\partial F_{p,q}}{\partial y} = \sum_{j=0}^{q-1} y^{q-1-j} x^p y^j.$$

For a one-ψ cyclic word $\mathrm{Tr}(\psi w)$ of bidegree (a, b), differentiating with respect to y gives one summand for each of the b occurrences of y in w, and differentiating with respect to x gives one summand for each of the a occurrences of x. Multiplying by the displayed cyclic derivatives of $F_{p,q}$ therefore produces pb positive one-ψ words and qa negative one-ψ words for each $w \in \mathsf{W}_{a,b}$, all of total bidegree $(a + p - 1, b + q - 1)$.

The bracket is a cycle because $F_{p,q}$ is invariant, hence $\{F_{p,q}, \mu_\epsilon\} = 0$ for every moment-map component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[x, y])$. Equivalently, the Hamiltonian vector field of $F_{p,q}$ preserves the Koszul differential $Q\psi = [x, y]$ and acts on Koszul homology. By Proposition 0.2.23, the target primitive one-ψ homology in that bidegree is one-dimensional. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ detects its coefficient. Since $\epsilon(\widetilde{\Psi}_{a,b}) = 1$, the averaged input gives

$$\epsilon\big(\{F_{p,q}, \widetilde{\Psi}_{a,b}\}\big) = pb - qa.$$

Also $\epsilon(\widetilde{\Psi}_{a+p-1,b+q-1}) = 1$. Hence the displayed formula follows. □

Corollary 0.2.39 *(Descendant action differs from the coadjoint action).* Under the boundary pairing $\delta_{a,b}[1] \mapsto [\Psi_{a,b}]$, the open Poisson action on polynomial one-ψ descendants differs from the closed coadjoint action on $\mathfrak{h}^\vee[1]$.

Proof. Proposition 0.2.38 gives

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\widetilde{\Psi}_{a+p-1,b+q-1}.$$

The closed coadjoint action is dual to the Poisson bracket. For $f = z_1^p z_2^q$ and the Taylor-dual basis vector $\delta_{a,b}$,

$$(\mathrm{ad}_f^\vee\, \delta_{a,b})(z_1^m z_2^n) = -\delta_{a,b}\big((pn - qm) z_1^{p+m-1} z_2^{q+n-1}\big).$$

Therefore, with the convention that negative indices give zero,

$$\mathrm{ad}^\vee_{z_1^p z_2^q}\, \delta_{a,b} = -(pb - qa + p - q)\delta_{a-p+1,b-q+1}.$$

For example, z_1 acts on open descendants by

$$\widetilde{\Psi}_{a,b} \longmapsto b\, \widetilde{\Psi}_{a,b-1},$$

whereas it acts coadjointly by

$$\delta_{a,b} \longmapsto -(b + 1)\delta_{a,b+1}.$$

The scalar edge $\{ \operatorname{Tr}(\phi_1), \widetilde{\Psi}_{0,1} \} = \Psi_{0,0}$ is removed with the scalar descendant. The theorem uses Corollary 0.2.25 for the PBW special-fibre product identification. A descendant deformation match requires a different boundary realization of the continuous dual, such as a Laurent or distributional model, together with its own bracket calculation. □

THEOREM 0.2.40 *(Classical bulk-to-boundary Hamiltonian map).* In the stable scalar-reduced Hamiltonian sector, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 0.2.21 stabilize to a linear map

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^0_{\mathrm{red}}(O^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}.$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra $\bar{A}$ to the scalar-reduced open trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\bar{A}}\big).$$

Proof. The map is well-defined on $\bar{A}$ by Corollary 0.2.27; constants map to the scalar N and are removed in the scalar-reduced sector. Lemma 0.2.36 gives the open reduced Poisson bracket. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 0.2.37 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \overline{(kn - lm)\operatorname{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

The closed Poisson bracket is

$$\{f, g\}_{\bar{A}} = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

and applying ∂_{bb} gives the same expression. Bilinearity proves the identity for all polynomials. □

COROLLARY 0.2.41 *(Classical local bulk-boundary coupling).* For scalar compactly supported test functions a, b on the brane line, define the smeared boundary Hamiltonian

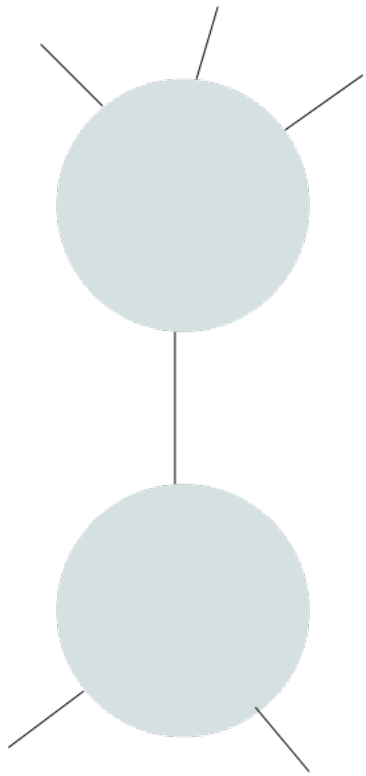
$$I_\partial(a, f) = \int_{\mathbb{R}} a(t)\, \overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\, dt$$

for $f \in \bar{A}$. In the local equal-time Poisson bracket convention,

$$\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}} = I_\partial(ab, \{f, g\}_{\bar{A}})$$

for $g \in \bar{A}$.

Proof. The test functions are scalar smearings of boundary observables. Form parts of the closed BV bracket are absent. The test functions have zero Poisson bracket with the matrix coordinates. The canonical bracket is local in t, so the equal-time delta-function multiplies the coefficients to $a(t)b(t)$, and the bracket of the inserted Hamiltonians is exactly the bracket computed in Theorem 0.2.40. Compact support removes boundary terms. □



The picture records this single canonical contraction.

Proof of Theorem 0.1.4. The open Hamiltonian-sector algebra is obtained from the product of the exact A-model self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ and the affine Koszul calculation for $\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m})$. Lemma 0.2.4 identifies the ghost-zero reduction with the gauged first-order matrix model

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Lemma 0.2.18 removes positive topological jets, and Lemma 0.2.26 identifies cyclic words with the same commutative monomial. Proposition 0.2.23 computes the primitive one-ψ homology, and Proposition 0.2.19 gives Proposition 0.2.28 after discarding the scalar classes:

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}} \cong \mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]).$$

On the closed side, Proposition 0.2.11 identifies the polynomial local Hamiltonian modes with $\bar{A}$, and their Taylor completion is $\mathfrak{h} = \widehat{\mathfrak{h}}$. The cotangent BF field supplies the shifted coadjoint copy, so the local Lie algebra is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$. Proposition 0.2.32 computes local observables as $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$. Proposition 0.2.35 gives the Koszul dual $\widehat{U(\mathfrak{g})}$, and the PBW/order Rees specialization gives $\widehat{\mathbf{S}}(\mathfrak{g})$ at $\hbar = 0$. Its degreewise polynomial subalgebra is $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$. Corollary 0.2.25 identifies $\mathfrak{h}^\vee$ with $\bar{A}_{\mathrm{desc}}$ as the nonconstant boundary-paired primitive one-ψ homology in the special fibre. Proposition 0.2.38 and Corollary 0.2.39 compute the open descendant bracket and separate it from the closed coadjoint

deformation. Finally Theorem 0.2.40 constructs the map from closed Hamiltonians to brane Hamiltonians and shows that it preserves the Poisson bracket. Corollary 0.2.41 packages this as the local classical bulk-boundary coupling. On monomials this gives the coefficient $(kn - lm)$ on both sides after scalar reduction. □

0.2.9.2 Formal Weyl/Moyal target, all orders

We use the Weyl/Moyal Rees convention for the formal quantization of the constant Poisson tensor:

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

Then the star commutator is odd in $\hbar$:

$$[f, g]_* = \hbar\{f, g\} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5),$$

where $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Thus the $\hbar^2$ coefficient vanishes in this quantization convention.

Proposition 0.2.42 *(Moyal coefficients on monomials).* Let $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$. For every $r \geq 0$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r},$$

where $(s)_q = s(s-1)\cdots(s-q+1)$ and $(s)_0 = 1$. Terms with a negative displayed exponent are understood to be zero; equivalently the corresponding falling factorial already vanishes. Consequently

$$[f, g]_* = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

so the second-order coefficient vanishes and the third-order coefficient is $\frac{1}{24} P^3(f, g)$.

Proof. Since

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1},$$

the two summands commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^{a} \otimes \partial_{z_1}^{a} \partial_{z_2}^{r-a}.$$

Applying this to $z_1^k z_2^l \otimes z_1^m z_2^n$ gives the displayed falling factorial formula. Interchanging f and g changes $P^r(f, g)$ by the sign $(-1)^r$, so the star commutator keeps exactly odd r and contributes the factor $2/(2^r r!)$ in order r. □

0.2.9.3 Third-order specialization and analytic graph problem

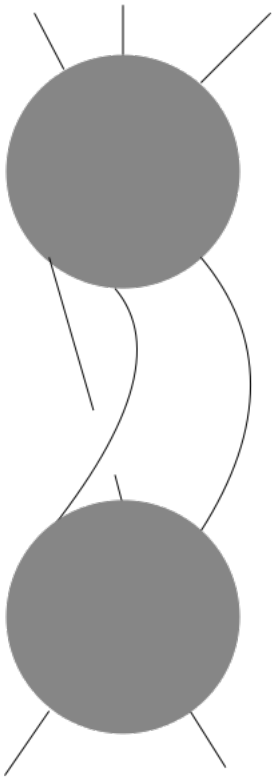
For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient from Proposition 0.2.42 is

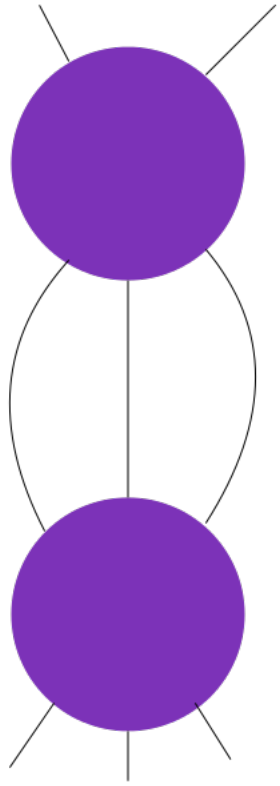
$$P^3(f, g) = \sum_{a=0}^{3} (-1)^a \binom{3}{a} (k)_{3-a} (l)_a (m)_a (n)_{3-a} z_1^{k+m-3} z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f, g]_* = \hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5).$$

The script `scripts/check_moyal_coefficients.py` checks this first-order coefficient, the vanishing of the second-order commutator coefficient, the displayed third-order formula, and the star-commutator normalisation on every monomial pair with exponents between 0 and 6, using exact rational arithmetic.





Problem 0.2.43 *(Analytic realization of the third-order graph coefficient).* Construct the closed-open Feynman model equipped with the analytic data of Problem 0.2.12. Assume the quantum master equation has no remaining anomaly in the Hamiltonian sector, and fix the propagator, bulk-boundary kernel, trace normalization, and large-N connected normalization compatibly with the first-order bracket. Prove that the connected third-order closed-open graph calculation realizes the algebraic coefficient

$$\frac{\hbar^3}{24} P^3(f, g)$$

in the Weyl/Moyal commutator. The formal algebraic coefficient is proved above. The analytic equality belongs to the Feynman-integral problem.

Remark 0.2.44. In characteristic zero, GL_N is linearly reductive on rational representations. Therefore taking GL_N-invariants is exact, and

$$H\big((\mathcal{R}_N)^{GL_N}, Q\big) \cong H(\mathcal{R}_N, Q)^{GL_N}.$$

This justifies the invariant calculation used above. A Chevalley BRST model for the quotient stack introduces Lie algebra ghosts as presentation data. Linearly reductive exactness identifies derived algebraic invariants with ordinary invariants in this setting.

Remark 0.2.45. The gravitational calculation uses string states, while the gauge-theory calculation uses a brane of the form a point cross a Lagrangian. A different B-brane replaces the associative algebra by a factorization algebra. With two topological directions the open algebra is an E_2 algebra, and the duality statement is the corresponding E_2 Koszul duality.

Remark 0.2.46. The same local method applies to D_0 branes on $\mathbb{R}^2 \times \mathbb{C}^4$.

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